



**BAULKHAM HILLS HIGH SCHOOL 2023
YEAR 11 ASSESSMENT TASK 1
MATHEMATICS ADVANCED**

MARKING COVER SHEET

FINAL MARK

NAME: _____ **Teacher:** _____

Page	Functions F1	Statistics S1	Total
Multiple Choice	Q2 -7, 9 /7	Q1, 8, 10 /3	/10
Page-8	Q11, 12a /6		/6
Page-9	Q12b, 13 /7		/7
Page-10	Q15 /3	Q14 /2	/5
Page-11	Q16, 17 /5		/5
Page-12	Q19 /2	Q18 /3	/5
Page-13	Q20, 21 /4		/4
Page-14	Q22, 23 /5		/5
Page-15	Q24 /5		/5
Page-16		Q25 /5	/5
Page-17	Q26, 27 /4		/4
Page-18	Q28 /4		/4
Page-19	Q29, 30 /4		/4
Page-20	Q31, 32 /5		/5
Page-21	Q34 /3	Q33 /2	/5
TOTAL	/64	/15	/79



BAULKHAM HILLS HIGH SCHOOL
YEAR 11 ASSESSMENT TASK 1, 2023
MATHEMATICS ADVANCED

NAME: _____

Teacher: _____

Mathematics

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct (arrow pointing to B)

- Start Here →
- | | | | | | | | |
|----------------------------|-------------------------|-------------------------|-------------------------|-----------------------------|-------------------------|-------------------------|-------------------------|
| 1. A ☐ | B ☐ | C ☐ | D ☐ | 6. A ☐ | B ☐ | C ☐ | D ☐ |
| 2. A ☐ | B ☐ | C ☐ | D ☐ | 7. A ☐ | B ☐ | C ☐ | D ☐ |
| 3. A ☐ | B ☐ | C ☐ | D ☐ | 8. A ☐ | B ☐ | C ☐ | D ☐ |
| 4. A ☐ | B ☐ | C ☐ | D ☐ | 9. A ☐ | B ☐ | C ☐ | D ☐ |
| 5. A ☐ | B ☐ | C ☐ | D ☐ | 10. A ☐ | B ☐ | C ☐ | D ☐ |



2023

PRELIMINARY ASSESSMENT TASK 1

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 34, show relevant mathematical reasoning and/or calculations

Total marks: 79

Section I – 10 marks (pages 4 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 69 marks (pages 8 – 21)

- Attempt Questions 11 – 34
- Allow about 1 hour and 15 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1** A fair 6-sided die is rolled twice. What is the probability of getting at least one 6?

- A. $\frac{1}{3}$
 - B. $\frac{1}{6}$
 - C. $\frac{11}{36}$
 - D. $\frac{5}{18}$
-

- 2** What is the minimum value of the function $y = 2x^2 + 6x + 2$?

- A. $\frac{-23}{2}$
- B. $\frac{-7}{2}$
- C. $\frac{-5}{2}$
- D. $\frac{31}{2}$

3 Which of the following functions is **NOT** a one-to-one function?

A. $f(x) = 4 - x^2, x \geq 0$

B. $f(x) = 1 - 4x$

C. $f(x) = \sqrt{4 - x^2}$

D. $f(x) = \frac{4}{x}$

4 Which of the following is **not** a polynomial?

A. $x^2 + 5x + 12$

B. $\frac{6}{x}(3x^2 + 2x)$

C. $(2x + 7)3x^{\frac{1}{2}}$

D. $2x(x^5 + 2x^3 - 4)^3$

5 What is $\frac{1}{4\sqrt[3]{(x+3)^5}}$ in index form?

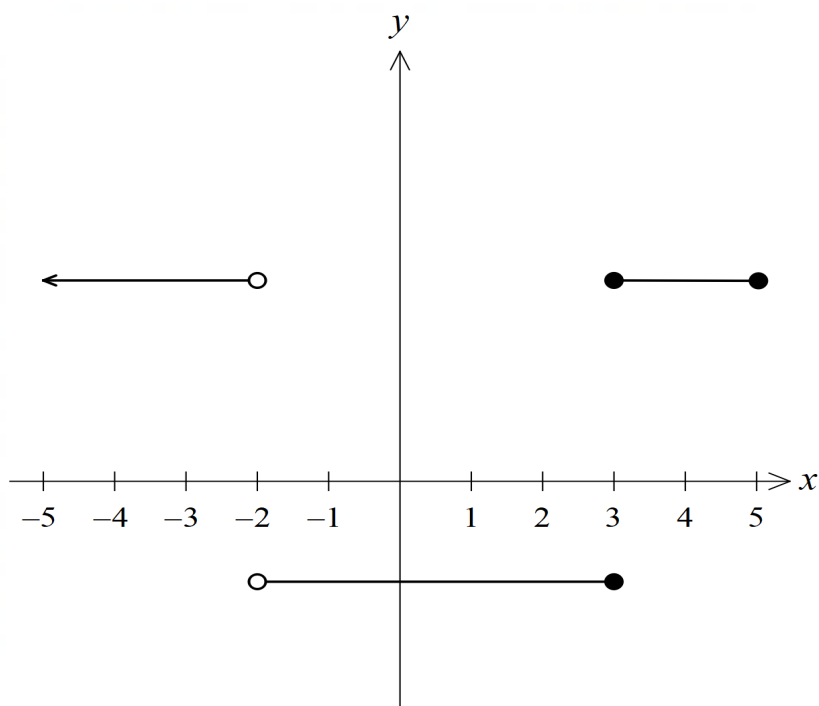
A. $4(x+3)^{\frac{-5}{3}}$

B. $4(x+3)^{\frac{5}{3}}$

C. $\frac{1}{4}(x+3)^{\frac{3}{5}}$

D. $\frac{1}{4}(x+3)^{\frac{-5}{3}}$

- 6 What is the domain of the graph drawn below?



- A. All real x .
B. $x \leq 5$, $x \neq -2$
C. $-4 > x \geq 5$
D. $x < 5$, $x \neq -2$
-
- 7 Which of the following is equivalent to $8x^3 - 10x^2 + 3x$?

- A. $x(4x + 3)(2x - 1)$
B. $x(4x - 3)(2x - 1)$
C. $x(4x - 3)(2x + 1)$
D. $x(4x + 3)(2x + 1)$
-

- 8 All of Bob's friends either jog or go to the gym. He knows that 70% of his friends jog, and 35% jog and go to a gym. What percentage of Bob's friends go to the gym, but do not jog?

A. 35%
B. 65%
C. 30%
D. 70%

- 9 What is the natural domain of $f(x) = \frac{1}{\sqrt{3-x}}$?

A. $(-\infty, 3]$
B. $(-\infty, 3)$
C. $[3, \infty)$
D. $(3, \infty)$

- 10 In a best of three sets tennis match (where the first player to win two sets wins the match), Chris has a probability of $\frac{2}{3}$ of winning a set. What is the probability that Chris wins the match?

A. $\frac{2}{3}$
B. $\frac{8}{9}$
C. $\frac{20}{27}$
D. $\frac{64}{81}$

End of Section I

Section II

69 marks

Attempt Questions 11–34

Allow about 1 hour and 15 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing paper is available. If you use extra paper, write your name, your teacher's name on the paper and clearly indicate which question you are answering.

Question 11 (4 marks)

- (a) Simplify $4\sqrt{12} - \sqrt{50} - \sqrt{27}$. 2

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- (b) Simplify $4y^{-2}(x)^{\frac{3}{2}}$, leaving your answer without fractional or negative indices. 2

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Question 12 (4 marks)

- (a) Factorise $x^3 - 64x$. 2

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- (b) Factorise $x^2 - 11xy - 1 + 11y$. 2

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Question 13 (5 marks)

Simplify the following:

- (a) $\frac{6}{x+4} + \frac{3x}{x^2+8x+16}$ 2

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- (b) $\frac{x^2-9}{x^2-2x-15} \div \frac{x^2-5x+6}{3x^2-12}$ 3

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Question 14 (2 marks)

Amy buys 4 tickets in a raffle in which 50 tickets are sold. The tickets are drawn for first and second prizes.

- (a) What is the probability that Amy wins the first prize? **1**

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- (b) What is the probability that Amy does not win the first prize but wins the second prize? **1**

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Question 15 (3 marks)

Given that $f(x) = x^2 + 2$ and $g(x) = 2x + 5$, find a simplified expression for:

- (a) $f(x) \times g(x)$ **1**

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- (b) $2g(a^2) - f(2a)$ **2**

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Question 16 (3 marks)

Boyle's Law is a gas law that states that the pressure **P** in a container is inversely proportional to the volume **V** of the container. At a given time, the volume of the container is 27.5 cm^3 and the pressure is 2.5 kg/cm^2 .

- (a) Find the pressure if the volume is reduced to 20 cm^3 . **2**

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- (b) Find the volume if the pressure is 2.75 kg/cm^2 . **1**

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Question 17 (2 marks)

A function is defined as $f(x) = \begin{cases} -x+8, & x < -8 \\ -x^2-3, & -8 \leq x < 4. \\ x^3, & x \geq 4 \end{cases}$ **2**

Evaluate $f(-3) + f(4)$.

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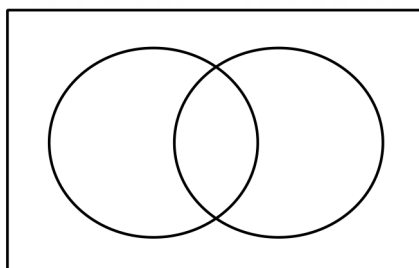
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Question 18 (3 marks)

In a group of 35 students, all either play sport or a musical instrument, 14 play both and 8 only play sport.

- (a) Complete the Venn diagram for the given information.

1



- (b) Find the probability that a student chosen at random will play a musical instrument given that the student plays sport.

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- (c) Find the probability that a student chosen at random will not play sport given that the student plays a musical instrument.

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Question 19 (2 marks)

Simplify $\frac{5^{2m} - 5^m}{5^m - 1}$.

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Question 20 (3 marks)

Solve the simultaneous equations $x + 3y = 2$ and $y = 2x + 11$.

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Question 21 (1 mark)

Alex solves a linear function simultaneously with a quadratic function to find any points of intersection, if they exist. Describe what happens geometrically if Alex finds that there is no solution.

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Question 22 (2 marks)

Solve $6 - 3x - 5x^2 = 0$ and give your answer correct to 3 decimal places.

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Question 23 (3 marks)

Rationalise the denominator $\frac{7 + \sqrt{5}}{\sqrt{5} - 1}$.

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Question 24 (5 marks)

- (a) Express $x^2 - 6x + 8$ in the form of $(x - h)^2 + k$. 2

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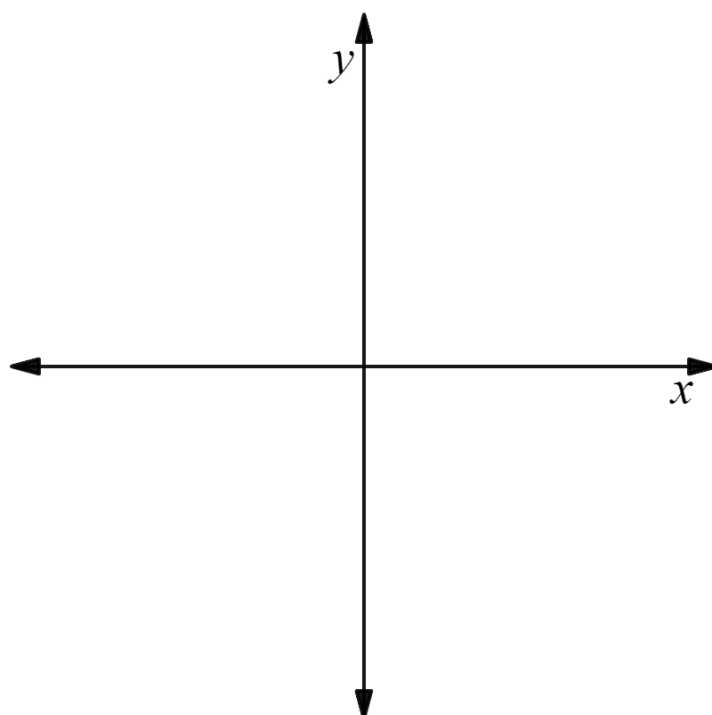
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- (b) Hence, sketch the quadratic function $f(x) = x^2 - 6x + 8$. Show all the important features of the graph. 2

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- (c) State the range of function $f(x) = x^2 - 6x + 8$ using interval notation. 1

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Question 25 (5 marks)

A box contains twelve chocolates, four of which are milk chocolate and eight are dark chocolate. Nathan eats three chocolates chosen randomly from the box.

- (a) Draw a probability tree diagram showing the above information. **1**

- (b) Find the probability that the first chocolate eaten is a dark chocolate. **1**

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- (c) Find the probability that Nathan eats three dark chocolates. **1**

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- (d) Find the probability that Nathan eats exactly one dark chocolate. **2**

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Question 26 (2 marks)

- (a) Find the discriminant of the quadratic equation $2x^2 - 9x + 7 = 0$. **1**

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- (b) Hence, determine whether the roots are rational or irrational. **1**

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Question 27 (2 marks)

- For what values of k does the quadratic equation $9x^2 + kx + 1 = 0$ have equal roots? **2**

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Question 28 (4 marks)

Find the domain and range of the following functions:

(a) $f(x) = \sqrt{25 - x^2}$ **2**

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(b) $g(x) = \frac{1}{x-1} + 2$ **2**

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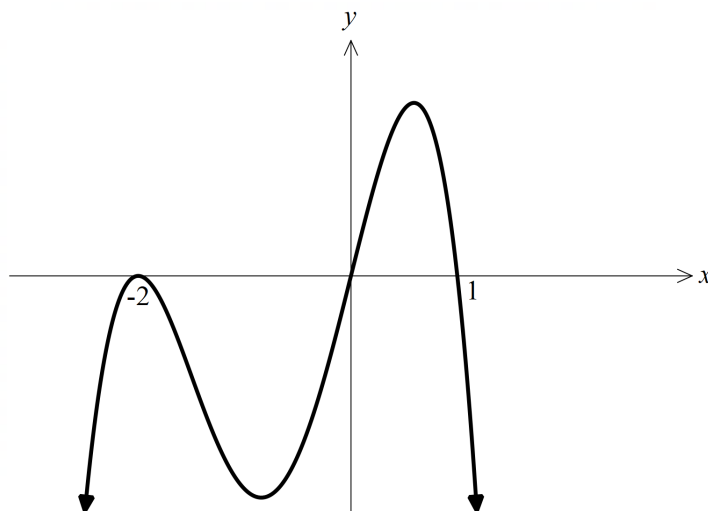
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Question 29 (1 mark)

The diagram below has leading coefficient either 1 or -1. Write down a possible equation of the polynomial of degree 4.

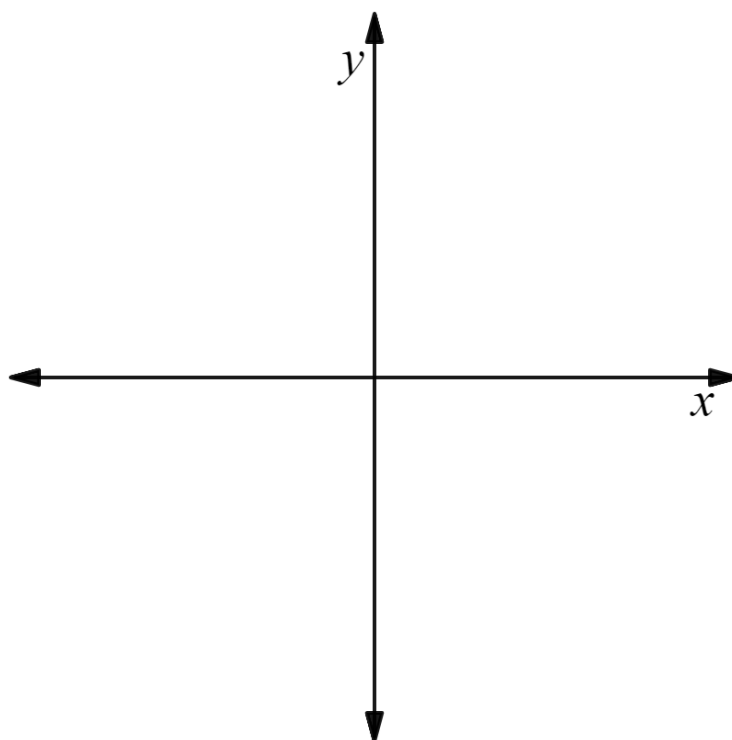
1



Question 30 (3 marks)

Sketch the graph of $y = -(x-3)(x^2-1)$, showing all important features.

3



Question 31 (3 marks)

The cost of producing x litres of orange juice is $(20 + x)$ dollars. If the income from the sale of x litres of the juice is $2x^2$ dollars, calculate the number of litres that must be sold to make a profit of \$800. **3**

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Question 32 (2 marks)

Solve $3^x + 3^{x+1} - 36 = 0$. **2**

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Question 33 (2 marks)

Given that $P(X) = 0.6$ and $P(X \cup Y) = 0.78$. Determine whether X and Y are independent if $P(Y) = 0.45$

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Question 34 (3 marks)

Solve $x - \sqrt{25 - x^2} = 1$.

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End of Paper

ADDITIONAL WORKING OUT SPACE:

If you are continuing a question on this page, write “Continued on extra page” at the end of the allocated space for the question you are continuing. Write the question number clearly before continuing/starting a question on this page.

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YEAR 11 ASSESSMENT TASK 1
MATHEMATICS ADVANCED**

MARKING COVER SHEET

FINAL MARK

NAME: _____ **Teacher:** _____

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Page-12	Q19 /2	Q18 /3	/5
Page-13	Q20, 21 /4		/4
Page-14	Q22, 23 /5		/5
Page-15	Q24 /5		/5
Page-16		Q25 /5	/5
Page-17	Q26, 27 /4		/4
Page-18	Q28 /4		/4
Page-19	Q29, 30 /4		/4
Page-20	Q31, 32 /5		/5
Page-21	Q34 /3	Q33 /2	/5
TOTAL	/64	/15	/79



BAULKHAM HILLS HIGH SCHOOL
YEAR 11 ASSESSMENT TASK 1, 2023
MATHEMATICS ADVANCED

NAME: _____

Teacher: _____

Mathematics

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Start
Here →

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|---|--|
| 1. A ☐ B ☐ C ☒ D ☐ | 6. A ☐ B ☒ C ☐ D ☐ |
| 2. A ☐ B ☐ C ☒ D ☐ | 7. A ☐ B ☒ C ☐ D ☐ |
| 3. A ☐ B ☐ C ☒ D ☐ | 8. A ☐ B ☐ C ☒ D ☐ |
| 4. A ☐ B ☐ C ☒ D ☐ | 9. A ☐ B ☒ C ☐ D ☐ |
| 5. A ☐ B ☐ C ☐ D ☒ | 10. A ☐ B ☐ C ☒ D ☐ |



2023

PRELIMINARY ASSESSMENT TASK 1

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
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Total marks: 79

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 A fair 6-sided die is rolled twice. What is the probability of getting at least one 6?

A. $\frac{1}{3}$

B. $\frac{1}{6}$

☒ C. $\frac{11}{36}$

D. $\frac{5}{18}$

$$1 - P(\text{No 6's})$$

$$1 - \frac{5}{6} \times \frac{5}{6}$$

$$1 - \frac{25}{36} = \frac{11}{36}$$

- 2 What is the minimum value of the function $y = 2x^2 + 6x + 2$?

A. $\frac{-23}{2}$

B. $\frac{-7}{2}$

☒ C. $\frac{-5}{2}$

D. $\frac{31}{2}$

$$2(x^2 + 3x + 1)$$

$$2\left[\left(x + \frac{3}{2}\right)^2 - \frac{5}{4}\right]$$

$$2\left(x + \frac{3}{2}\right)^2 - \frac{5}{2}$$

$$\therefore \text{Vertex } \left(-\frac{3}{2}, -\frac{5}{2}\right)$$

$$\therefore \text{Minimum value is } -\frac{5}{2}$$

3 Which of the following functions is **NOT** a one-to-one function?

A. $f(x) = 4 - x^2, x \geq 0$

B. $f(x) = 1 - 4x$

☒ C. $f(x) = \sqrt{4 - x^2}$ *It doesn't satisfy the horizontal line test.*

D. $f(x) = \frac{4}{x}$

4 Which of the following is **not** a polynomial?

A. $x^2 + 5x + 12$

B. $\frac{6}{x}(3x^2 + 2x)$

☒ C. $(2x + 7)3x^{\frac{1}{2}}$ *as the power of x is in fraction*

D. $2x(x^5 + 2x^3 - 4)^3$

5 What is $\frac{1}{4\sqrt[3]{(x+3)^5}}$ in index form?

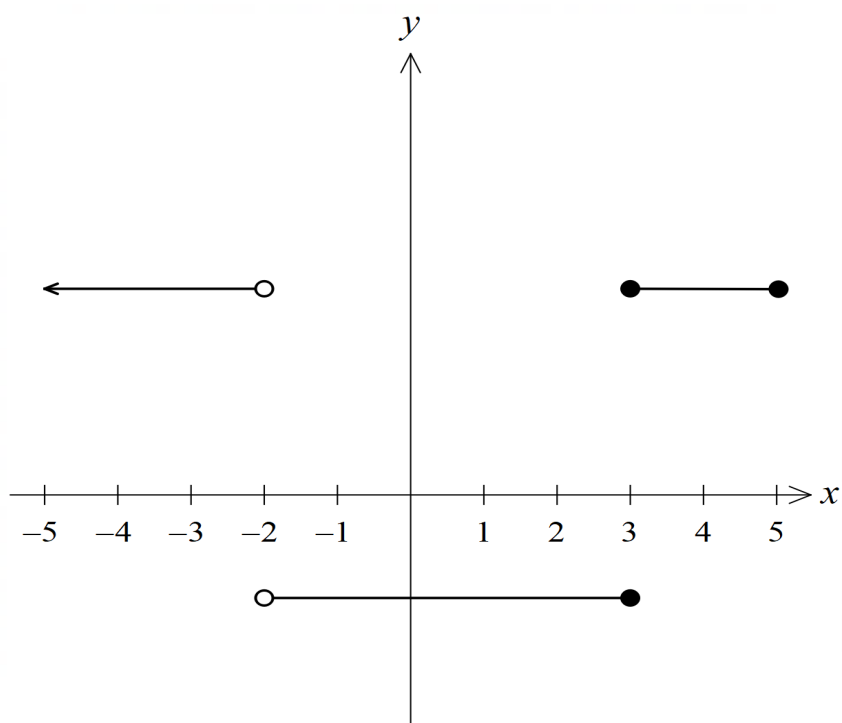
A. $4(x+3)^{\frac{-5}{3}}$

B. $4(x+3)^{\frac{5}{3}}$

C. $\frac{1}{4}(x+3)^{\frac{3}{5}}$

☒ D. $\frac{1}{4}(x+3)^{\frac{-5}{3}}$

- 6 What is the domain of the graph drawn below?



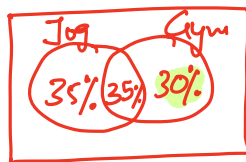
- A. All real x .
B. $x \leq 5, x \neq -2$
C. $-4 > x \geq 5$
D. $x < 5, x \neq -2$

-
- 7 Which of the following is equivalent to $8x^3 - 10x^2 + 3x$?

- A. $x(4x+3)(2x-1)$
B. $x(4x-3)(2x-1)$
C. $x(4x-3)(2x+1)$
D. $x(4x+3)(2x+1)$

$$\begin{aligned} & x(8x^2 - 10x + 3) \\ & x(8x^2 - 6x - 4x + 3) \\ & x(2x(4x - 3) - 1(4x + 3)) \\ & x(4x - 3)(2x - 1) \end{aligned}$$

- 8 All of Bob's friends either jog or go to the gym. He knows that 70% of his friends jog, and 35% jog and go to a gym. What percentage of Bob's friends go to the gym, but do not jog?



- A. 35%
B. 65%
C. 30%
D. 70%

- 9 What is the natural domain of $f(x) = \frac{1}{\sqrt{3-x}}$?

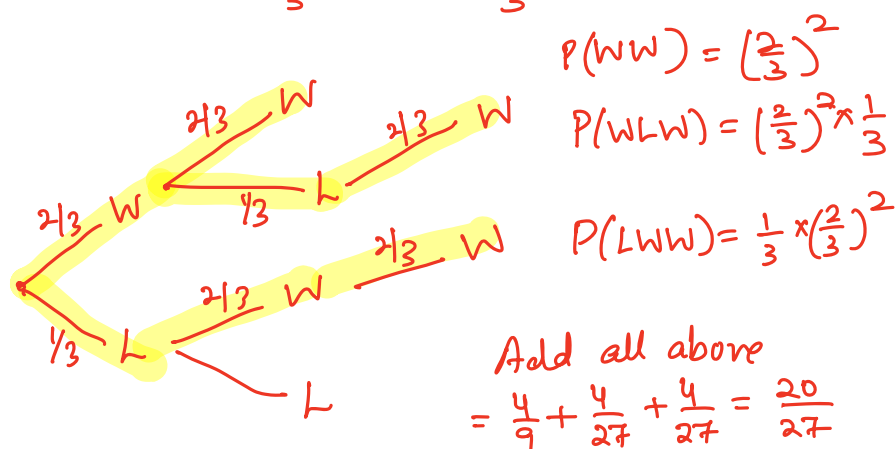
- A. $(-\infty, 3]$
B. $(-\infty, 3)$
C. $[3, \infty)$
D. $(3, \infty)$

$$\begin{aligned} 3 - x &> 0 \\ x &< 3 \\ (-\infty, 3) \end{aligned}$$

- 10 In a best of three sets tennis match (where the first player to win two sets wins the match), Chris has a probability of $\frac{2}{3}$ of winning a set. What is the probability that Chris wins the match?

$$P(W) = \frac{2}{3}, \quad P(L) = \frac{1}{3}$$

- A. $\frac{2}{3}$
B. $\frac{8}{9}$
C. $\frac{20}{27}$
D. $\frac{64}{81}$



End of Section I

Section II

69 marks

Attempt Questions 11–34

Allow about 1 hour and 15 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing paper is available. If you use extra paper, write your name, your teacher's name on the paper and clearly indicate which question you are answering.

Question 11 (4 marks)

(a) Simplify $4\sqrt{12} - \sqrt{50} - \sqrt{27}$.

2

$$\begin{aligned} & 8\sqrt{3} - 5\sqrt{2} - 3\sqrt{3} \\ & 5\sqrt{3} - 5\sqrt{2} \end{aligned}$$

1 mark for correctly decompose two surds
2 marks correct solution.

(b) Simplify $4y^{-2}(x)^{\frac{3}{2}}$, leaving your answer without fractional or negative indices. 2

$$\frac{4x\sqrt{x}}{y^2}$$

1 mark for correctly changing either negative indices to positive or $(x)^{\frac{3}{2}}$ to $x\sqrt{x}$
2 marks correct solution.

Question 12 (4 marks)

(a) Factorise $x^3 - 64x$.

2

$$\begin{aligned} & x(x^2 - 64) \\ & x(x - 8)(x + 8) \end{aligned}$$

1 mark for one correct factor
2 marks all correct factors

(b) Factorise $x^2 - 11xy - 1 + 11y$.

2

$$\begin{aligned}
 & x^2 - 1 - 11xy + 11y \\
 & (x^2 - 1) - 11y(x - 1) \\
 & (x - 1)(x + 1 - 11y)
 \end{aligned}$$

1 mark for correctly using grouping in pairs
2 marks for correct factors

Question 13 (5 marks)

Simplify the following:

(a) $\frac{6}{x+4} + \frac{3x}{x^2+8x+16}$

2

$$\begin{aligned}
 & \frac{6}{x+4} + \frac{3x}{(x+4)^2} \\
 & \frac{6(x+4) + 3x}{(x+4)^2} \\
 & \frac{9x + 24}{(x+4)^2}
 \end{aligned}$$

1 mark for correctly changing two fractions to a single fraction with common denominator.
2 marks correct solution.

(b) $\frac{x^2-9}{x^2-2x-15} \div \frac{x^2-5x+6}{3x^2-12}$

3

$$\begin{aligned}
 & \frac{(x-3)(x+3)}{(x-5)(x+3)} \times \frac{3(x^2-4)}{(x-2)(x-3)} \\
 & \frac{3(x+2)}{x-5} \\
 \text{or} & \frac{3x+6}{x-5}
 \end{aligned}$$

1 mark for factorising at least two expressions.
2 marks for factorising three expressions and changing $\div$ into $\times$.
3 marks correct solution

Question 14 (2 marks)

Amy buys 4 tickets in a raffle in which 50 tickets are sold. The tickets are drawn for first and second prizes.

- (a) What is the probability that Amy wins the first prize?

1

$$\frac{4}{50} = \frac{2}{25} \quad \text{1 mark for correct solution}$$

- (b) What is the probability that Amy does not win the first prize but wins the second prize?

1

$$\frac{46}{50} \times \frac{4}{49} = \frac{92}{1225} \quad \text{1 mark for correct solution}$$

Question 15 (3 marks)

Given that $f(x) = x^2 + 2$ and $g(x) = 2x + 5$, find a simplified expression for:

- (a) $f(x) \times g(x)$

1

$$(x^2 + 2)(2x + 5) \\ 2x^3 + 5x^2 + 4x + 10 \quad \text{1 mark for correct expression}$$

- (b) $2g(a^2) - f(2a)$

2

$$2(2a^2 + 5) - ((2a)^2 + 2) \\ 4a^2 + 10 - 4a^2 - 2 \\ 8$$

1 mark for either expressing correct $g(a^2)$ or $f(2a)$
2 marks for correct answer

Question 16 (3 marks)

Boyle's Law is a gas law that states that the pressure **P** in a container is inversely proportional to the volume **V** of the container. At a given time, the volume of the container is 27.5 cm^3 and the pressure is 2.5 kg/cm^2 .

- (a) Find the pressure if the volume is reduced to 20 cm^3 .

2

$$\begin{aligned}
 P &\propto \frac{1}{V} \Rightarrow P = \frac{k}{V} \\
 k &= PV = 27.5 \times 2.5 = 68.75 \quad \checkmark \\
 P &= \frac{k}{V} = \frac{68.75}{20} \\
 &= 3.4375 \text{ kg/cm}^2 \quad \checkmark
 \end{aligned}$$

1 mark to find k.
2 marks for both k and P.

- (b) Find the volume if the pressure is 2.75 kg/cm^2 .

1

$$V = \frac{k}{P} = \frac{68.75}{2.75} = 25 \text{ cm}^3$$

1 mark for correct solution.

Question 17 (2 marks)

A function is defined as $f(x) = \begin{cases} -x+8, & x < -8 \\ -x^2-3, & -8 \leq x < 4 \\ x^3, & x \geq 4 \end{cases}$

2

Evaluate $f(-3) + f(4)$.

$$\begin{aligned}
 &-(-3)^2 - 3 + 4^3 \\
 &-12 + 64 \\
 &52
 \end{aligned}$$

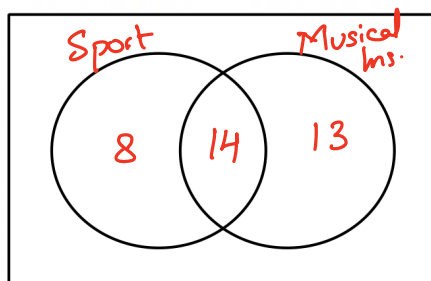
1 mark for either correct $f(-3)$ or $f(4)$
2 marks for correct value.

Question 18 (3 marks)

In a group of 35 students, all either play sport or a musical instrument, 14 play both and 8 only play sport.

- (a) Complete the Venn diagram for the given information.

1



1 mark to complete the Venn diagram correctly with labelled Sport and Musical instrument.

- (b) Find the probability that a student chosen at random will play a musical instrument given that the student plays sport.

1

$\frac{14}{22}$ or $\frac{7}{11}$ 1 mark for correct solution.

- (c) Find the probability that a student chosen at random will not play sport given that the student plays a musical instrument.

1

$\frac{13}{27}$ 1 mark for correct solution.

Question 19 (2 marks)

Simplify $\frac{5^{2m} - 5^m}{5^m - 1}$.

2

$$\frac{5^m (5^m - 1)}{5^m - 1}$$

$$5^m$$

1 mark to correctly factorise the numerator.
2 marks for correct solution.

Question 20 (3 marks)Solve the simultaneous equations $x + 3y = 2$ and $y = 2x + 11$.

3

$$x + 3y = 2$$

$$x + 3(2x + 11) = 2$$

$$x + 6x + 33 = 2$$

$$7x = -31$$

$$x = -\frac{31}{7}$$

$$y = 2\left(-\frac{31}{7}\right) + 11$$

$$= \frac{-62 + 77}{7}$$

$$= \frac{15}{7}$$

$$\therefore x = -\frac{31}{7}, y = \frac{15}{7}$$

1 mark for correctly using substitution or elimination method to solve simultaneous equations.

2 marks for correctly getting either x or y value.

3 marks for correct sol.

Question 21 (1 mark)

Alex solves a linear function simultaneously with a quadratic function to find any points of intersection, if they exist. Describe what happens geometrically if Alex finds that there is no solution.

1

For no solution, the line and quadratic function doesn't have any point of intersection.

1 mark for correct explanation.

Question 22 (2 marks)Solve $6 - 3x - 5x^2 = 0$ and give your answer correct to 3 decimal places.

2

$$-5x^2 - 3x + 6 = 0$$

$$x = \frac{3 \pm \sqrt{9 + 120}}{-10}$$

$$= \frac{3 + \sqrt{129}}{-10}, \quad \frac{3 - \sqrt{129}}{-10}$$

$$= -1.436, \quad 0.836$$

1 mark for correctly using quad. formula/ completing sq. method to solve the equation

2 marks with correct solution including 3 decimal places.

Question 23 (3 marks)Rationalise the denominator $\frac{7 + \sqrt{5}}{\sqrt{5} - 1}$.

3

$$\frac{7 + \sqrt{5}}{\sqrt{5} - 1} \times \frac{\sqrt{5} + 1}{\sqrt{5} + 1}$$

$$\frac{(7 + \sqrt{5})(\sqrt{5} + 1)}{5 - 1}$$

$$\frac{8\sqrt{5} + 12}{4}$$

$$2\sqrt{5} + 3$$

1 mark for multiplying and divide by $(\sqrt{5} + 1)$

2 marks for multiplying by the conjugate surd and simplifying either num. or den.

3 marks for correct solution in simplified form.

Question 24 (5 marks)

- (a) Express $x^2 - 6x + 8$ in the form of $(x-h)^2 + k$.

2

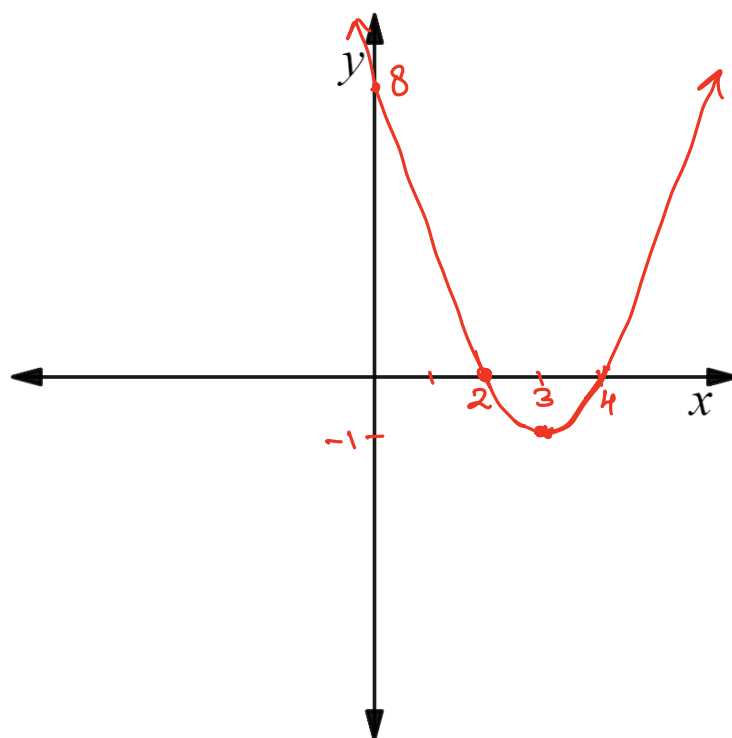
$$\begin{aligned} & x^2 - 6x + 8 \\ & x^2 - 6x + 9 - 1 \\ & (x-3)^2 - 1 \end{aligned}$$

1 mark for either correct h or k.
2 marks for correctly expressing in given form

- (b) Hence, sketch the quadratic function $f(x) = x^2 - 6x + 8$. Show all the important features of the graph.

2

$$\begin{aligned} & \text{y-intercept: } x=0, y=8 \quad (0, 8) \\ & \text{x-intercept: } y=0 \Rightarrow x^2 - 6x + 8 = 0 \\ & \quad (x-4)(x-2) = 0 \Rightarrow x=2, 4. \end{aligned}$$



1 mark for correct shape.
1 mark for correct x & y intercepts.
(vertex needs to be shown)

- (c) State the range of function $f(x) = x^2 - 6x + 8$ using interval notation.

1

$$\text{Range: } y \in [-1, \infty)$$

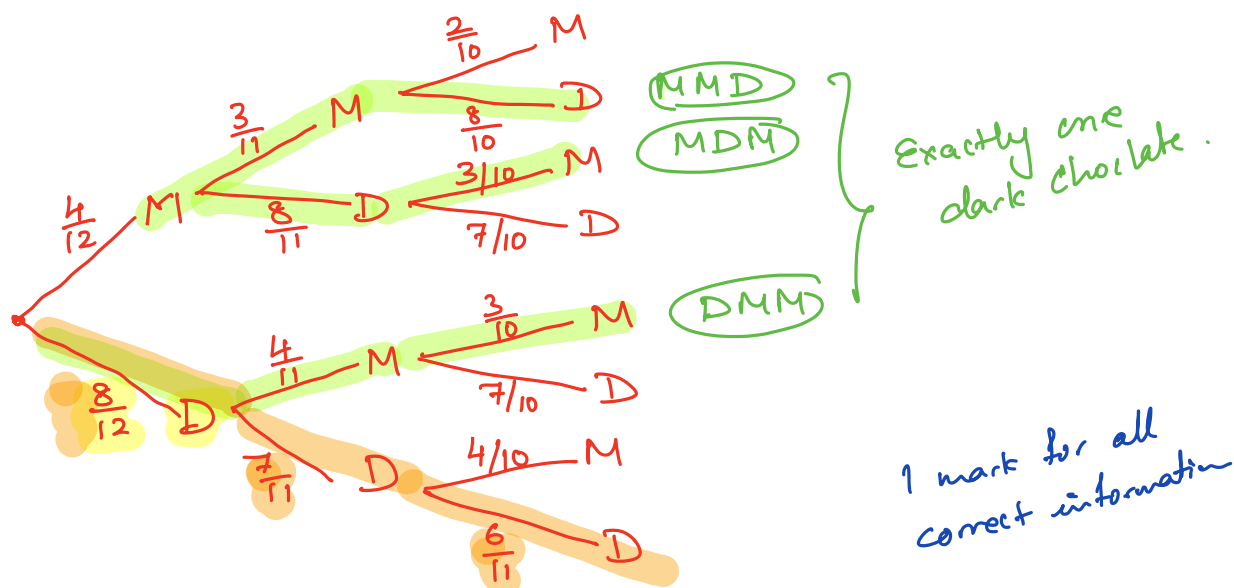
1 mark for correct interval notation

Question 25 (5 marks)

A box contains twelve chocolates, four of which are Milk chocolate and eight are Dark chocolate. Nathan eats three chocolates chosen randomly from the box.

- (a) Draw a probability tree diagram showing the above information.

1



- (b) Find the probability that the first chocolate eaten is a dark chocolate.

1

$\frac{8}{12} = \frac{2}{3}$ 1 mark for correct sol.

- (c) Find the probability that Nathan eats three dark chocolates.

1

$\frac{8}{12} \times \frac{7}{11} \times \frac{6}{10} = \frac{14}{55}$ 1 mark for correct sol.

- (d) Find the probability that Nathan eats exactly one dark chocolate.

2

$\left(\frac{8}{12} \times \frac{4}{11} \times \frac{3}{10} \right) \times 3 = \frac{12}{55}$ 1 mark for one correct e.g. MMD or MDM or DMM

2 marks for correct solution.

Question 26 (2 marks)

- (a) Find the discriminant of the quadratic equation $2x^2 - 9x + 7 = 0$.

1

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (-9)^2 - 4(2)(7) \\ &= 81 - 56 \\ &= 25\end{aligned}$$

1 mark for correct
discriminant

- (b) Hence, determine whether the roots are rational or irrational.

1

Since discriminant is perfect square
 $\therefore$ roots are rational.

1 mark for correct
answer with
explanation

Question 27 (2 marks)

For what values of k does the quadratic equation $9x^2 + kx + 1 = 0$ have equal roots?

2

$$\begin{aligned}\text{For equal roots: } \Delta &= 0 \\ b^2 - 4ac &= 0 \\ k^2 - 36 &= 0 \\ k &= \pm 6\end{aligned}$$

1 mark for finding the
discriminant and equating
it to zero.
2 marks for correct solution.

Question 28 (4 marks)

Find the domain and range of the following functions:

(a) $f(x) = \sqrt{25 - x^2}$

2

Domain: $x \in [-5, 5]$ or $-5 \leq x \leq 5$

Range: $y \in [0, 5]$ or $0 \leq y \leq 5$

1 mark for correct domain

1 mark for correct range.

(b) $g(x) = \frac{1}{x-1} + 2$

2

Domain: All real x except $x=1$

or $x \in \mathbb{R} \setminus \{1\}$ or $x \in (-\infty, 1) \cup (1, \infty)$

Range: All real y except $y=2$

or $y \in \mathbb{R} \setminus \{2\}$ or $y \in (-\infty, 2) \cup (2, \infty)$

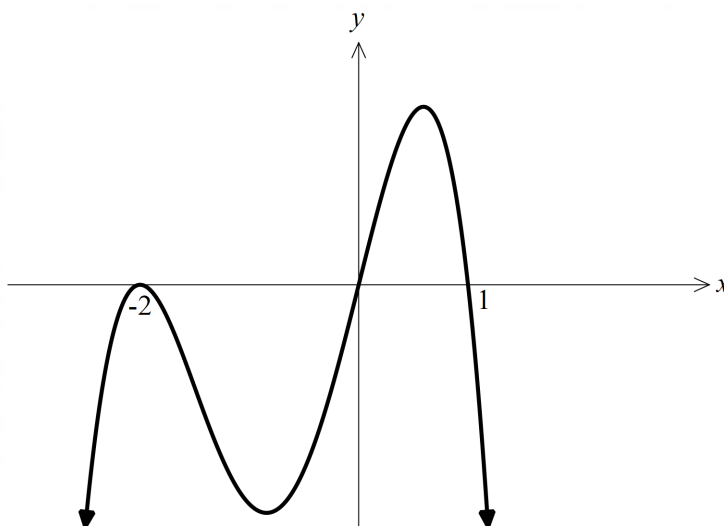
1 mark for correct domain

1 mark for correct range.

Question 29 (1 mark)

The diagram below has leading coefficient either 1 or -1. Write down a possible equation of the polynomial of degree 4.

1



$y = -x(x+2)^2(x-1)$

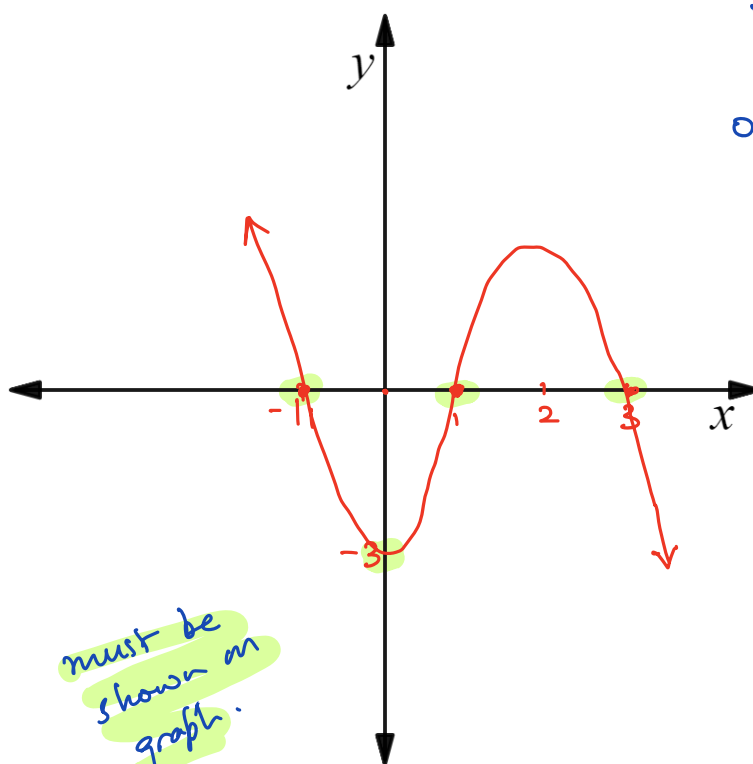
1 mark for correct equation. e.g $y = \dots$
 $P(x) = \dots$

Question 30 (3 marks)

Sketch the graph of $y = -(x-3)(x^2-1)$, showing all important features.

3

$= -(x-3)(x-1)(x+1)$



1 mark for shape
or
or any 2 correct x-int.
2 marks for at least
two correct x-intercepts
including shape.
3 marks for correct
graph (shape, intercepts)

must be shown on graph.

Question 31 (3 marks)

The cost of producing x litres of orange juice is $(20 + x)$ dollars. If the income from the sale of x litres of the juice is $2x^2$ dollars, calculate the number of litres that must be sold to make a profit of \$800. 3

$$\begin{aligned}
 &\text{Profit} = \text{Selling Price} - \text{Cost Price} \\
 &800 = 2x^2 - 20 - x \\
 &2x^2 - x - 820 = 0 \\
 &x = \frac{1 \pm \sqrt{1 + 6560}}{4} \\
 &= \frac{1 \pm 81}{4} \\
 &= 20.5, -20 \quad (\text{Reject -ve}) \\
 &\therefore 20.5 \text{ litres must be sold to make a profit of \$800.}
 \end{aligned}$$

{ 1 mark for correct expression of profit
 2 marks for solving quad. eq.
 3 marks for correct sol.

Question 32 (2 marks)

Solve $3^x + 3^{x+1} - 36 = 0$. 2

$$\begin{aligned}
 &3^x + 3 \cdot 3^x = 36 \\
 &4 \cdot 3^x = 36 \\
 &3^x = 9 \\
 &3^x = 3^2 \\
 &x = 2
 \end{aligned}$$

{ 1 mark for changing $3^{x+1} = 3 \cdot 3^x$
 or $3^x + 3^{x+1} = 4 \cdot 3^x$
 2 marks for correct solution.

Question 33 (2 marks)

Given that $P(X) = 0.6$ and $P(X \cup Y) = 0.78$. Determine whether X and Y are independent if $P(Y) = 0.45$

2

$$\begin{aligned}
 P(X \cup Y) &= P(X) + P(Y) - P(X \cap Y) \\
 0.78 &= 0.6 + 0.45 - P(X \cap Y) \\
 P(X \cap Y) &= 0.27 \\
 P(X) \times P(Y) &= 0.6 \times 0.45 = 0.27 \\
 \text{Since } P(X \cap Y) &= P(X) \times P(Y) \\
 \therefore X \text{ and } Y &\text{ are independent.}
 \end{aligned}$$

1 mark to find $P(X \cap Y)$
2 marks for correct solution and showing X and Y are independent

Question 34 (3 marks)

Solve $x - \sqrt{25 - x^2} = 1$.

3

$$\begin{aligned}
 x - 1 &= \sqrt{25 - x^2} \quad \text{--- (x)} \\
 \text{Square both sides.} \\
 (x - 1)^2 &= 25 - x^2 \\
 x^2 - 2x + 1 &= 25 - x^2 \\
 2x^2 - 2x - 24 &= 0 \\
 x^2 - x - 12 &= 0 \\
 (x - 4)(x + 3) &= 0 \\
 x = 4, -3 &\quad \text{(Reject -3 as it won't satisfy)} \\
 x = 4 &\text{ is the only solution}
 \end{aligned}$$

1 mark to create a quad. eq. by rearranging the terms.
2 mark to solve the quad. eq. correctly.
3 marks for correct sol.

End of Paper

If you are continuing a question on this page, write “Continued on extra page” at the end of the allocated space for the question you are continuing. Write the question number clearly before continuing/starting a question on this page.

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